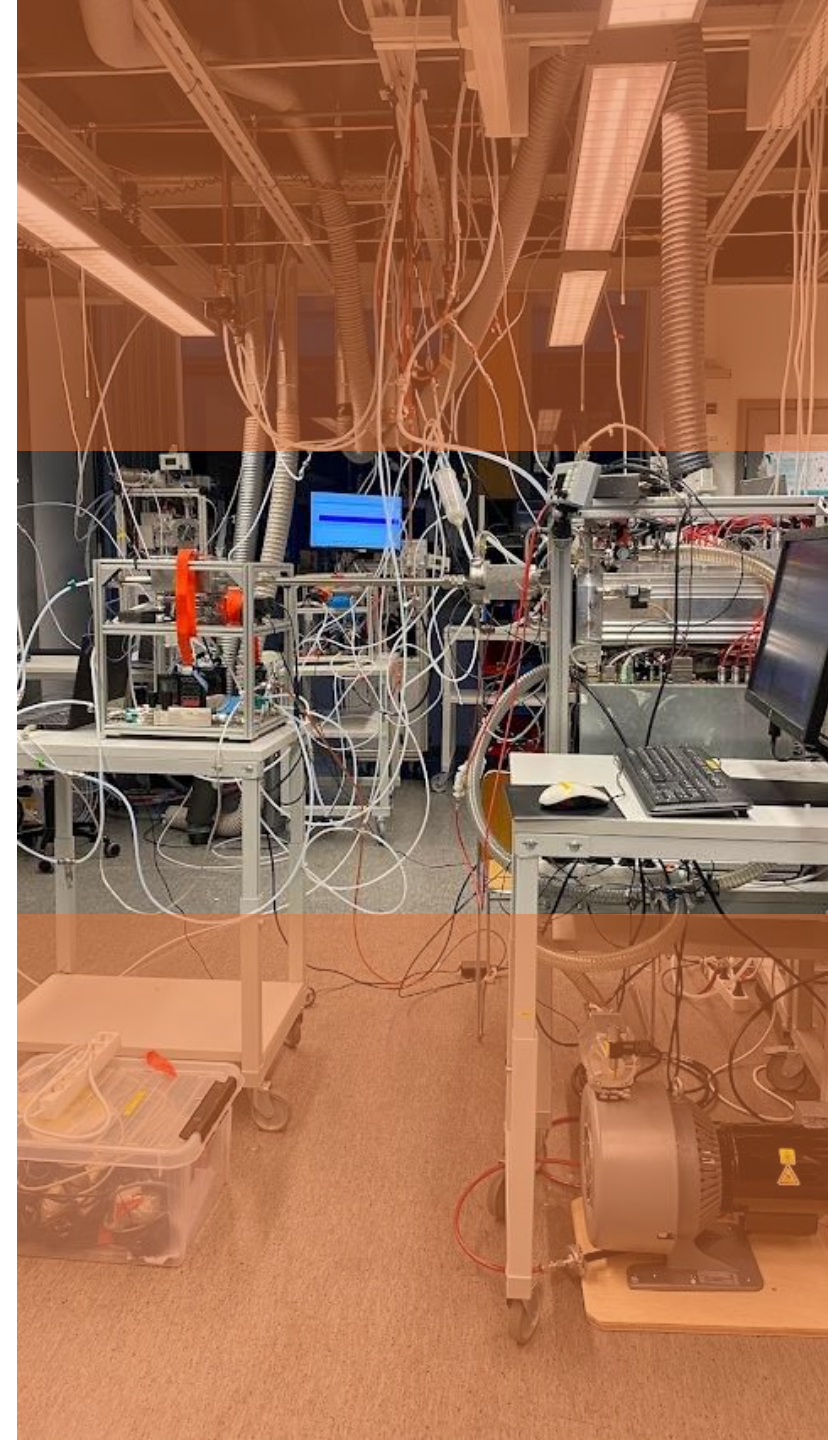


Sulfuric acid calibration of CI-API-ToF

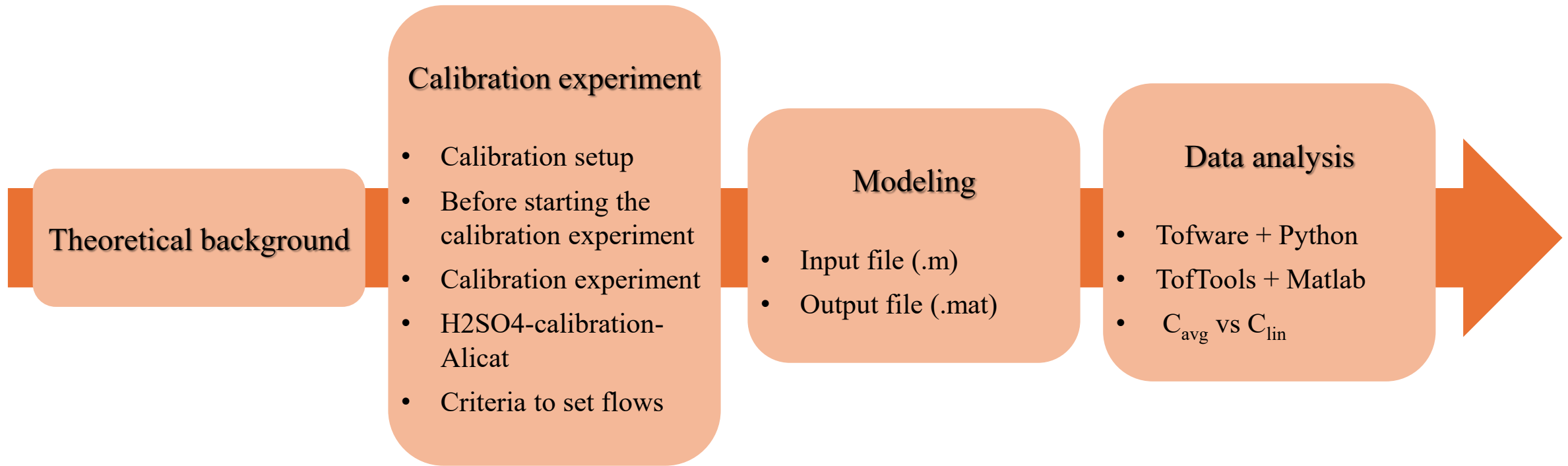
Operational instructions for the ACTRIS CiGas – UHEL calibration setup

Cecilia Righi, Nina Sarnela

Latest update: May 2026



Outline



Theoretical background

- ❑ The Analog-to-Digital Converter in the Time-of-Flight mass analyzer generates a mass spectrum in which signal intensity (typically ions/s) is plotted as a function of the mass-to-charge ratio (m/z).
- ❑ To obtain **quantitative** information, we must determine how the measured signal intensity relates to a known analyte concentration. In other words, we need a **calibration**.
- ❑ **Condensable vapors** readily condense onto surfaces due to their low volatility. Hence, they cannot be simply stored in gas tanks, and producing well-defined concentrations of these low-volatility vapors is not straightforward.
- ❑ A reliable and widely used method to calibrate CI-API-ToF, first described by Kürten *et al.* (2012), uses **sulfuric acid as calibrant**. The calibration was originally developed for **NO_3^- CIMS**, as it works particularly well with this reagent-ion chemistry.



THE JOURNAL OF
PHYSICAL CHEMISTRY A

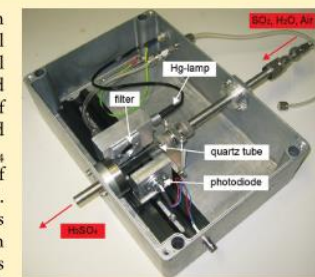
Article
pubs.acs.org/JPCA

Calibration of a Chemical Ionization Mass Spectrometer for the Measurement of Gaseous Sulfuric Acid

Andreas Kürten,* Linda Rondo, Sebastian Ehrhart, and Joachim Curtius

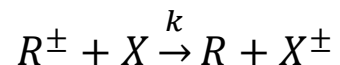
Institute for Atmospheric and Environmental Sciences, Goethe-University Frankfurt am Main, Altenhöferallee 1, 60438 Frankfurt am Main, Germany

ABSTRACT: The accurate measurement of the gaseous sulfuric acid concentration is crucial within many fields of atmospheric science. Instruments utilizing chemical ionization mass spectrometry (CIMS) measuring H_2SO_4 , therefore, require a careful calibration. We have set up a calibration source that can provide a stable and adjustable concentration of H_2SO_4 . The calibration system initiates the production of sulfuric acid through the oxidation of SO_2 by OH. The hydroxyl radical is produced by UV photolysis of water vapor. A numerical model calculates the H_2SO_4 concentration provided at the outlet of the calibration source. From comparison of this concentration and the signals measured by CIMS, a calibration factor is derived. This factor is evaluated to be $1.1 \times 10^{10} \text{ cm}^{-3}$, which is in good agreement with values found in the literature for other CIMS instruments measuring H_2SO_4 . The calibration system is described in detail and the results are discussed. Because the setup is external to the CIMS instrument, it offers the possibility for future CIMS intercomparison measurements by providing defined and stable concentrations of sulfuric acid.



Theoretical background

General



$$\frac{d[R^{\pm}]}{dt} = -k[R^{\pm}][X]$$

$$[X] \stackrel{\star}{=} \frac{1}{kt} \cdot \ln \left(1 + \frac{[X^{\pm}]}{[R^{\pm}]} \right) \stackrel{\star}{\approx} \frac{1}{kt} \cdot \frac{[X^{\pm}]}{[R^{\pm}]}$$

$$[X] \approx \frac{1}{kt} \cdot \frac{1}{T_X} \cdot \underbrace{\frac{i(X^{\pm})}{i(R^{\pm})}}_{S_X}$$

★ assuming $[X]$ as a constant over t

★ assuming $[X^{\pm}] \ll [R^{\pm}]$

Sulfuric acid

$$[H_2SO_4] = C_{SA} \cdot S_{H_2SO_4}$$

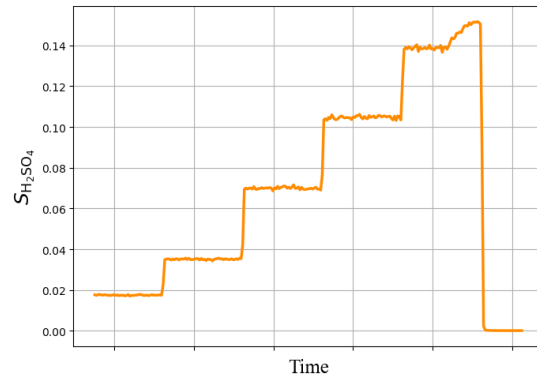
$$S_{H_2SO_4} = \frac{HSO_4^- + HNO_3 \cdot HSO_4^- + H_2SO_4 \cdot HSO_4^- + (H_2SO_4)_2 \cdot HSO_4^-}{NO_3^- + HNO_3 \cdot NO_3^- + (HNO_3)_2 \cdot NO_3^- + H_2O \cdot NO_3^-}$$

$$C_{SA} = \frac{[H_2SO_4]}{S_{H_2SO_4}}$$

Heinritzi *et al.*, 2016

Kürten *et al.*, 2012

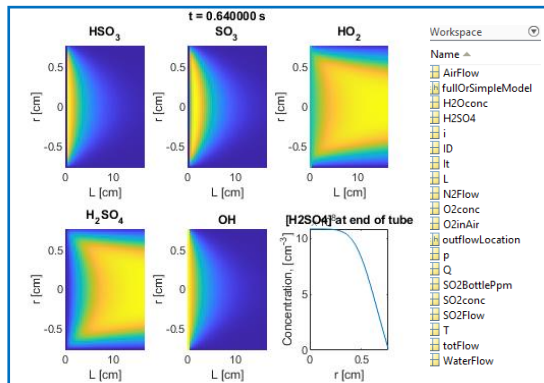
Theoretical background



Sulfuric acid

$$[H_2SO_4] = C_{SA} \cdot S_{H_2SO_4}$$

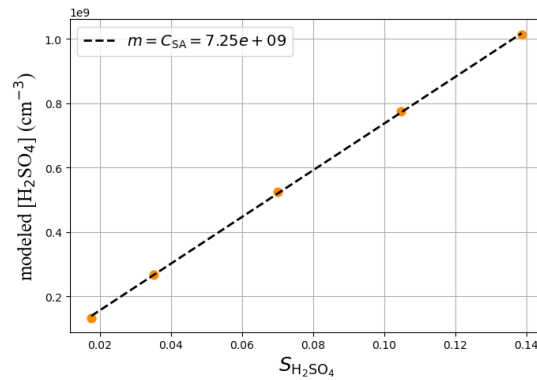
$$S_{H_2SO_4} = \frac{HSO_4^- + HNO_3 \cdot HSO_4^- + H_2SO_4 \cdot HSO_4^- + (H_2SO_4)_2 \cdot HSO_4^-}{NO_3^- + HNO_3 \cdot NO_3^- + (HNO_3)_2 \cdot NO_3^- + H_2O \cdot NO_3^-}$$



modeled

$$C_{SA} = \frac{[H_2SO_4]}{S_{H_2SO_4}}$$

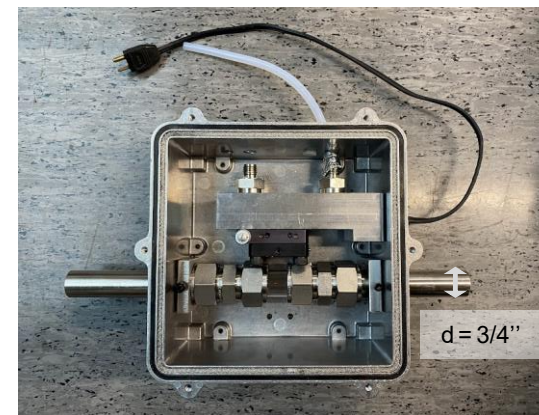
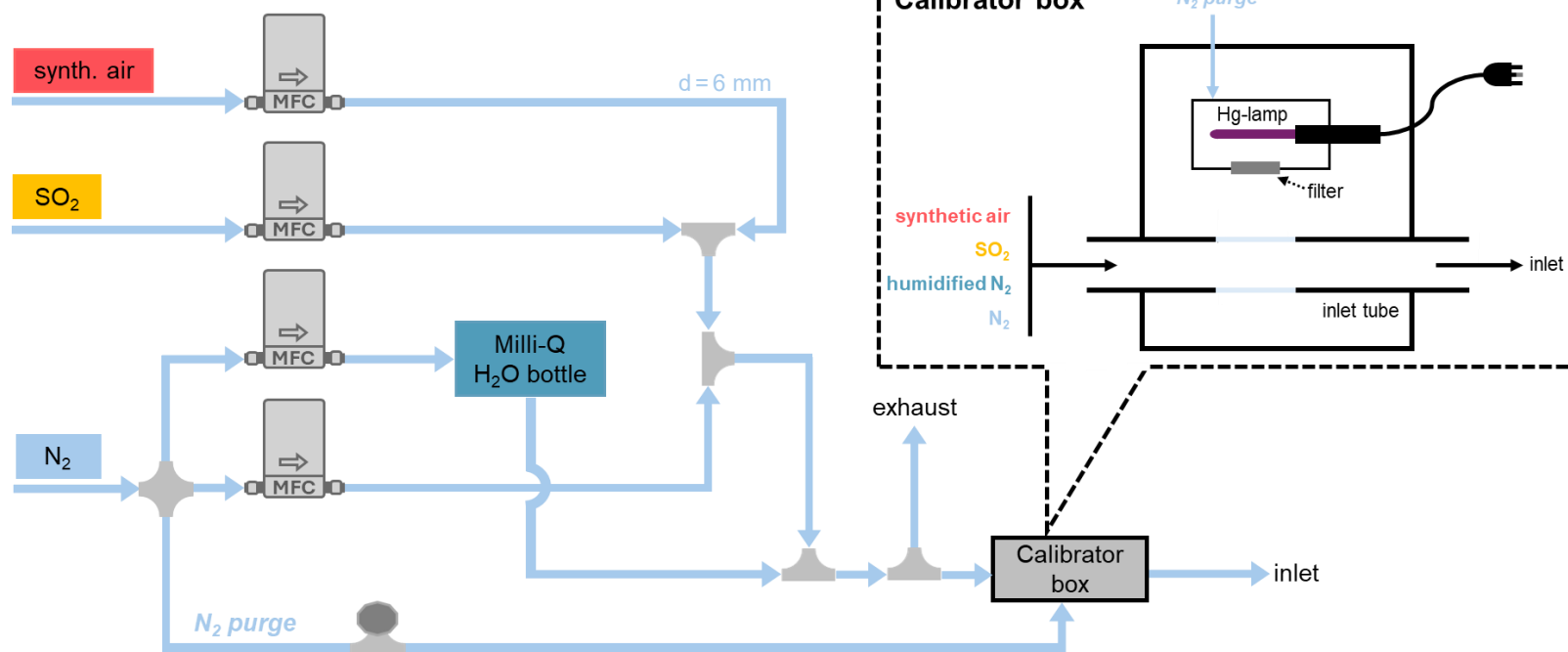
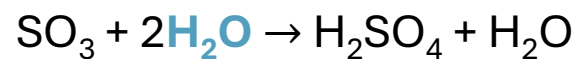
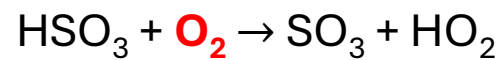
experiment



Kürten *et al.*, 2012

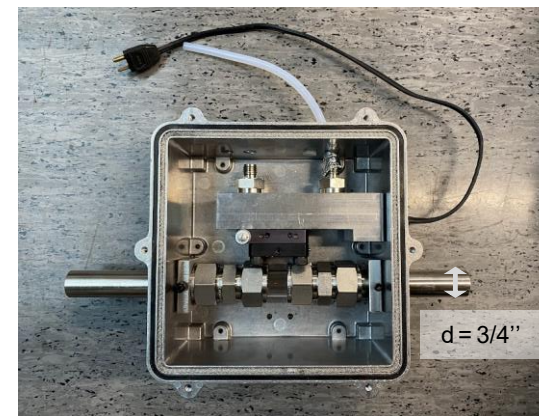
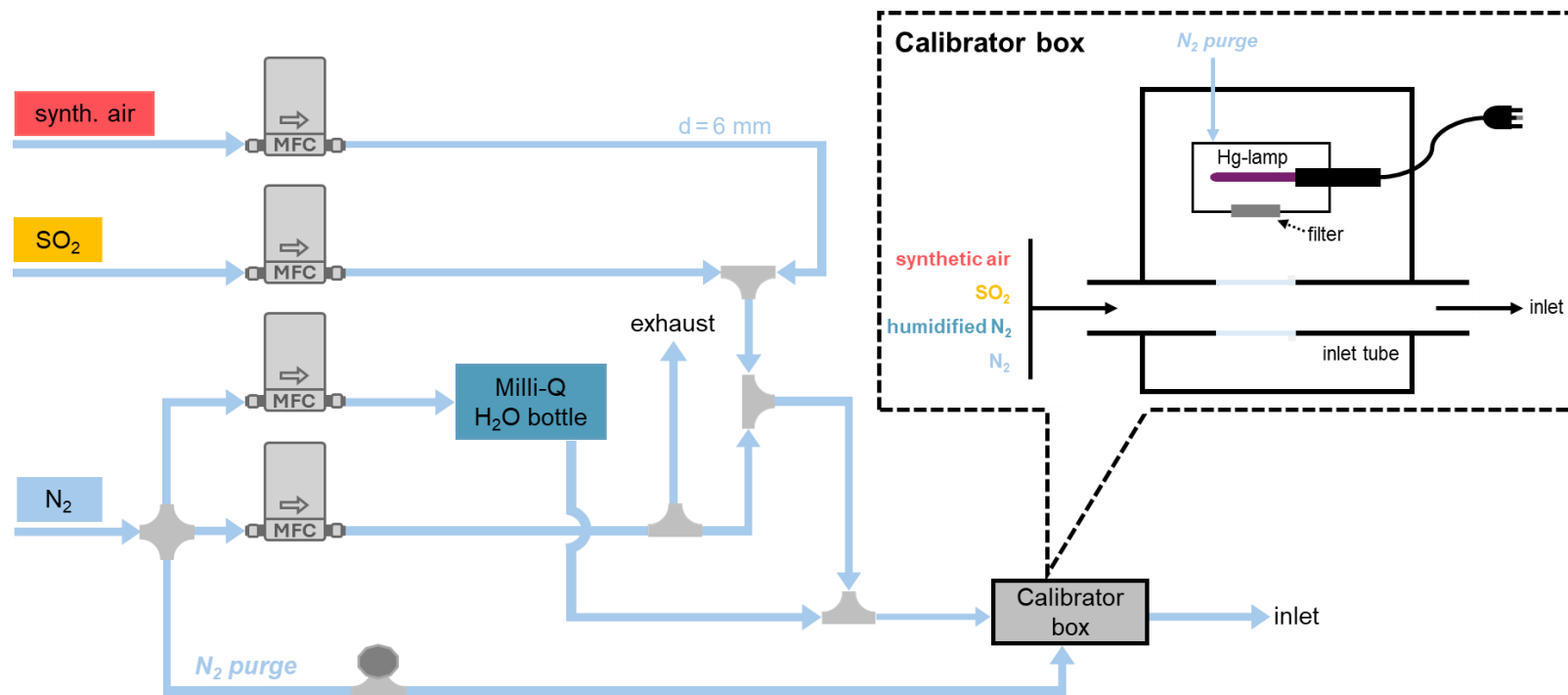
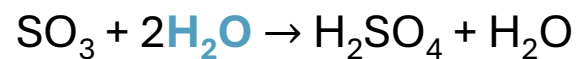
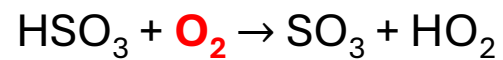
Calibration experiment

Calibration setup A



Calibration experiment

Calibration setup B

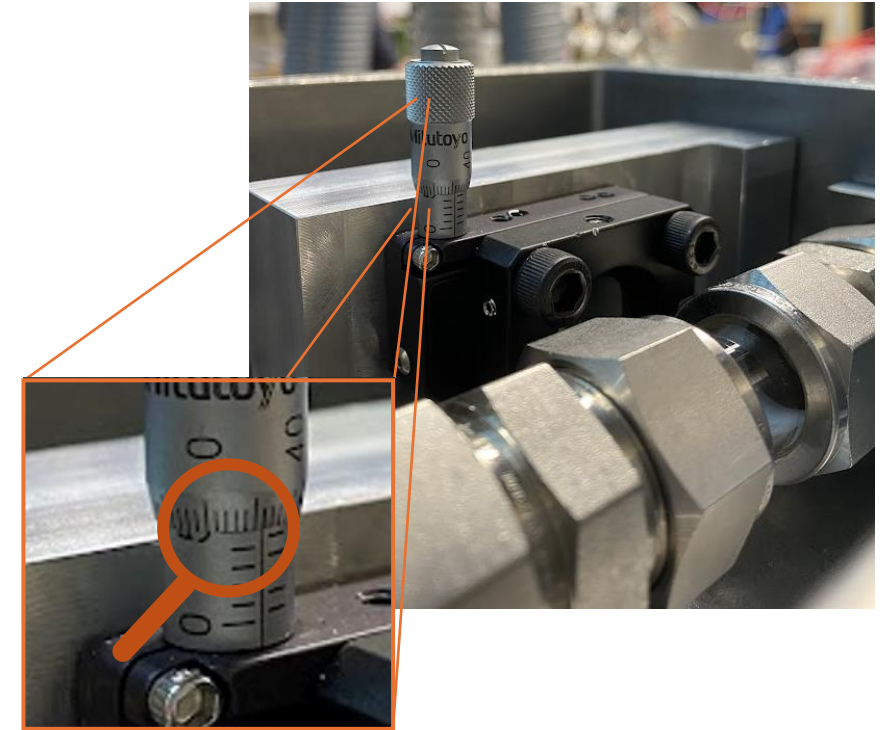


Calibration experiment

Before starting the calibration experiment

- ❑ Ensure that the calibrator box is in good condition, especially if it has been moved several times before the experiment:
 - there should be no crack in the quartz tube;
 - the filter should be set to the same position used in the actinometry experiment (5mm for this setup).
- ❑ Measure the inlet flow (Q) of CI-API-ToF and note it down.
- ❑ Connect the calibration setup to CI-API-ToF and record the dimensions of the inlet tube (ID^* and L^{**}). **Do not connect the exhaust line at this stage** (let it sample ambient air).
- ❑ Measure the N_2 flush flow to be connected to the calibrator box. Adjust it to 1.5 L min^{-1} using the needle valve.
- ❑ **Only after the N_2 flush flow is properly set, turn on the Hg-lamp.** The lamp must warm up for at least 90 minutes before use.

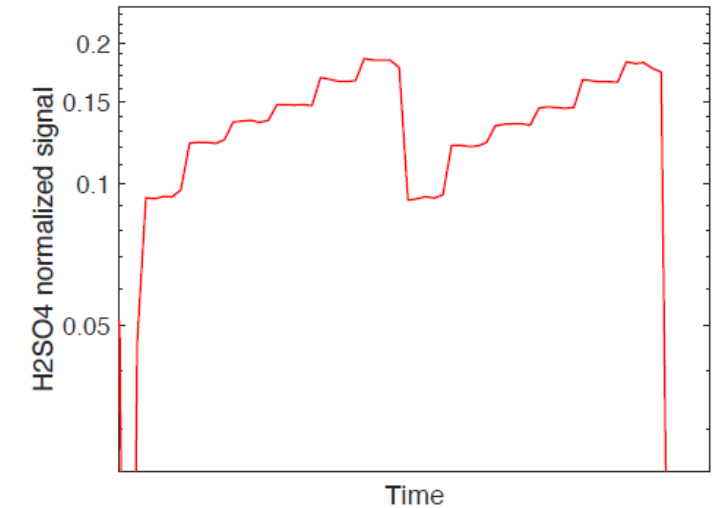
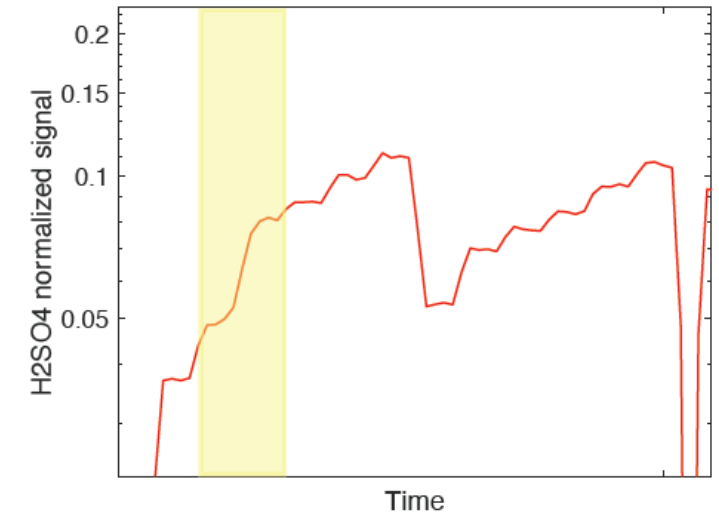
- * ID refers to the inner diameter of the inlet tube. For a 3/4" stainless steel tube, it is usually 0.62", which corresponds to approximately 15.6 mm.
- ** L refers to the distance traveled by the H_2SO_4 produced inside the calibrator box before reacting with the reagent ion in the CI source. Hence, measure from approximately the middle of the calibrator box (where the quartz tube is located) to the point where the reagent ion meets the sample gas in the IMR region of the CI inlet (for Eisele inlet, measure up to the black line; for MION inlet, measure up to the middle of the ion source).



Calibration experiment

Calibration experiment

- ❑ When the Hg-lamp is ready for the experiment, **connect the exhaust line** and set the flow rates on the MFCs. This can be done directly on the MFC displays or by using the Python scripts located in the folder *H2SO4-calibration-Alicat* (see the following slide for more details).
- ❑ We recommend performing a calibration experiment consisting of at least five 10-minute steps by varying the N_2 flow passing through the Milli-Q H_2O bottle and repeating the full calibration sequence twice in a row. This will help assess whether the lamp remains stable throughout the calibration.
- ❑ **Before starting the experiment**, it is good practice to **inject only SO_2 for a few minutes** (the required time depends on the length of the SO_2 line): since the SO_2 flow is relatively small compared to the other flows, it takes longer to fill the tubing, and you may lose H_2SO_4 that could form during the first calibration step.



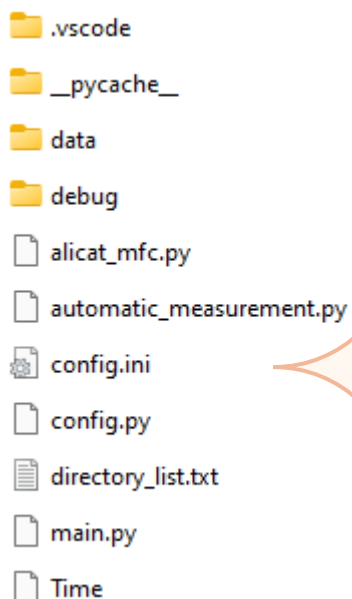
(top) Hg-lamp unstable during the first calibration
(bottom) Hg-lamp stable throughout the experiment

Righi, 2024

Calibration experiment

H2SO4-calibration-Alicat

- 1 Adjust the flow settings as needed in *config.ini*



```
# General notes on editing this file:
# - Comments must be on their own line. E.g. 'device_id = Dev1 # Qwerty' doesn't work (Maybe there is a way to get it working)
# - Section names probably allows whitespace but I wanted to avoid any potential problems by the naming convention currently used

# MFC's serial port settings
[MFC:Serial_port]
port = COM4
baudrate = 19200
bytesize = 8
parity = N
stopbits = 1
timeout = 1
# Software flow control
xonxoff = 0
# Hardware flow control
rtscts = 0
# individual addresses. NOTE: each MFC has individual address! (A = N2, B = N2 wet, C = SO2, D = synth. air.)
address = A,B,C,D

# Scaling settings for the MFCs
[MFC:Scaling]
# Flow
f_span = 1,1,1,1
f_offset = 0,0,0,0
# Temperature
t_span = 1,1,1,1
t_offset = 0,0,0,0
# Pressure
p_span = 68.9476,68.9476,68.9476,68.9476
p_offset = 0,0,0,0

# Automatic measurement loop settings
[Automatic_measurement]
# Measurement frequency [Hz]
freq = 1

# Set time intervals (in seconds) to change the flow rates. NOTE: these are indeed time intervals Delta_t, not time from starting the script.
time_intervals = 600,600,600,600,600
# Setting flow rate [sLPM or sccm, depends on the MFC] for each MFC. Symbol "--" separates different time intervals.
set_flow_init = 10,0.125,0.003,0.07--10,0.25,0.003,0.07--10,0.5,0.003,0.07--10,0.75,0.003,0.07--10,1,0.003,0.07
```

COM port corresponding to the connection to the MFCs (check in *Device Manager*)

Verify and, if needed, correct the individual MFC addresses shown on the MFC displays

- 2 Run *main.py* from the terminal

Set each step duration and corresponding flows following the instructions

Calibration experiment

Criteria to set flows

The set of flows is considered optimal for the calibration experiment on a specific CI-API-ToF when:

- ❑ The **total injected flow is equal to or greater than Q** ; otherwise, the instrument will draw the remaining flow from the exhaust line. 
- ❑ The **H₂SO₄ concentrations do not saturate the reagent ion**; the reagent ion must remain clearly the most abundant species throughout the calibration for the principles of chemical ionization to hold.
- ❑ The H₂SO₄ concentrations produced at each step are sufficiently high for the calibration factor to remain constant. According to Kürten *et al.* (2012), this requires **concentrations above 3×10^7 molecules cm⁻³**.

We have identified sets of flows that generally work well (for setup A) when the CI-API-ToF samples at $\sim 10 \text{ L min}^{-1}$ (Eisele-type inlet) and when it samples at $\sim 20 \text{ L min}^{-1}$ (MION inlet):

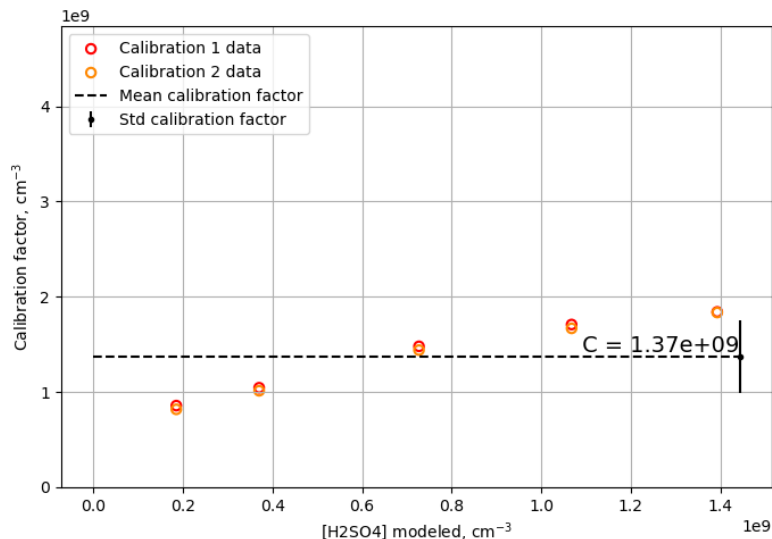
Q (lpm)	N2Flow (lpm)	WaterFlow (mlpm)	SO2Flow (mlpm)	AirFlow (mlpm)
10	10	125, 250, 500, 750, 1000	3	70
20	20	750, 1000, 1250, 1500, 1750	3	70

Calibration experiment

Criteria to set flows

The set of flows is considered optimal for the calibration experiment on a specific CI-API-ToF when:

- ❑ The **total injected flow is equal to or greater than Q** ; otherwise, the instrument will draw the remaining flow from the exhaust line.
 inlet flow
- ❑ The **H₂SO₄ concentrations do not saturate the reagent ion**; the reagent ion must remain clearly the most abundant species throughout the calibration for the principles of chemical ionization to hold.
- ❑ The H₂SO₄ concentrations produced at each step are sufficiently high for the calibration factor to remain constant. According to Kürten *et al.* (2012), this requires **concentrations above 3×10^7 molecules cm⁻³**.



If the calibration factors calculated show a **clear increasing trend** with increasing H₂SO₄ concentration, the **WaterFlow should be increased** from the lowest calibration steps and scaled accordingly.

For example, instead of using [125, 250, 500, 750, 1000], try shifting the range upward to [500, 750, 1000, 1250, 1500].

Calibration experiment

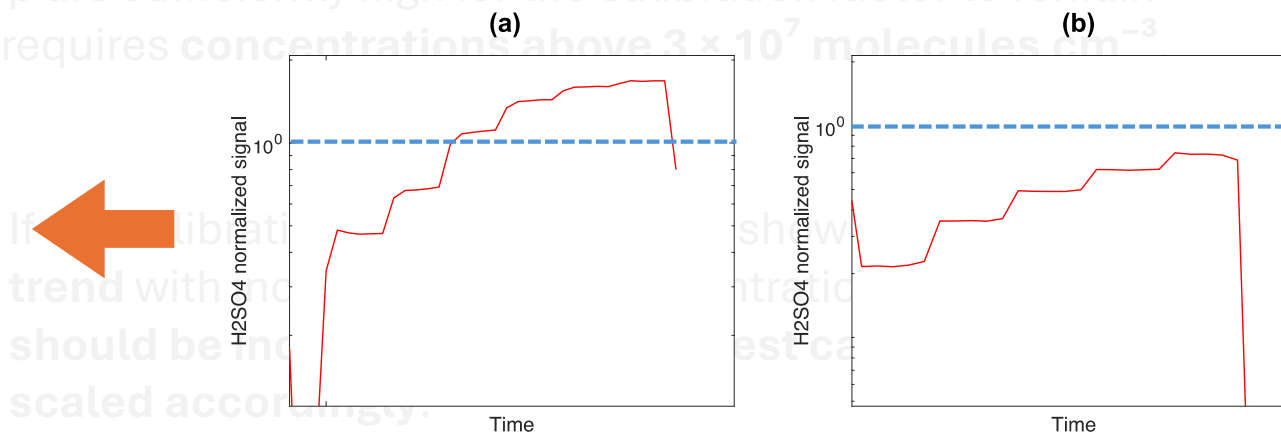
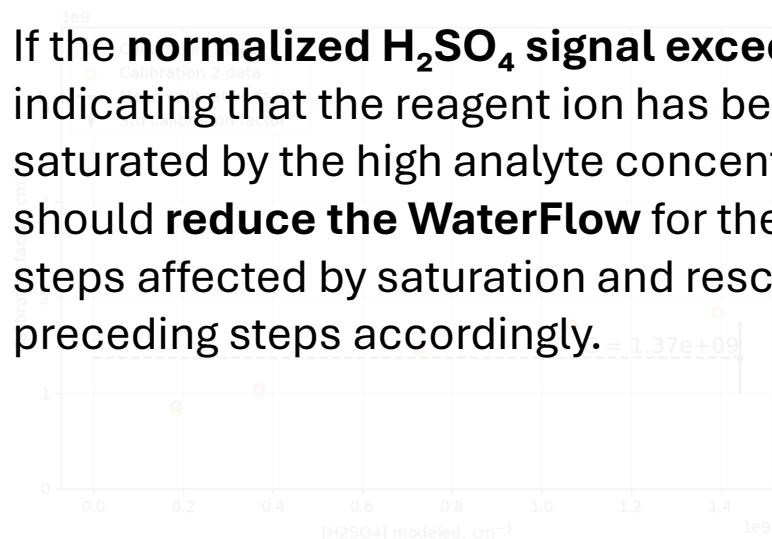
Criteria to set flows

The set of flows is considered optimal for the calibration experiment on a specific CI-API-ToF when:

- ❑ The **total injected flow is equal to or greater than Q** ; otherwise, the instrument will draw the remaining flow from the exhaust line. inlet flow
- ❑ The **H_2SO_4 concentrations do not saturate the reagent ion**; the reagent ion must remain clearly the most abundant species throughout the calibration for the principles of chemical ionization to hold.

❑ The H_2SO_4 concentrations produced at each step are sufficiently high for the calibration factor to remain constant. According to Kürten *et al.* (2012), this requires concentrations above 3×10^7 molecules cm^{-3} .

If the **normalized H_2SO_4 signal exceeds 1**, indicating that the reagent ion has been saturated by the high analyte concentration, you should **reduce the WaterFlow** for the calibration steps affected by saturation and rescale the preceding steps accordingly.



(a) Time series showing a calibration affected by reagent-ion saturation starting from the third step of the calibration experiment.
(b) Time series of a properly executed calibration, with the reagent ion remaining clearly the most abundant species across all calibration steps.

Righi, 2024

Modeling

Input file (m.)

- 1 Adjust the input parameters as needed in *Input_Parameters.m*

cmd_calib1Matlab.m
diff_sa_rh.m
gormleyKennedy.m
Input_Parameters.m
odesolveMatlab.m
vappresw.m

- 2 Run *Input_Parameters.m* in Matlab

```
%% input parameters
% set these inputs and run the file
clear all;

T = 296.76; % K % adapted
p = 98400; % Pa % adapted

ID = 15.60; % mm, inner diameter tube % adapted
L = 840.00; % mm % adapted
Q = 10.12; % lpm, not slpm % adapted

It = 5.91e11; % at Q flow rate % adapted

N2Flow = 10.0; % slpm % adapted
AirFlow = 70; % smlpm % adapted
WaterFlow = [125 250 500 750 1000]'; % smlpm % adapted
SO2Flow = 3; % smlpm % adapted

SO2BottlePpm = 5000; % ppm % adapted

O2inAir = 0.209;

outflowLocation = 'after'; % outflow tube located before or after injecting air, water, and so2
fullOrSimpleModel = 'full'; % simple: Gormley&Kennedy approximation, full: flow model (much slower)

%% computation begins

if strcmp(outflowLocation,'after')
    totFlow = N2Flow+AirFlow/1000+WaterFlow/1000+SO2Flow/1000;
else
    totFlow = Q*ones(size(WaterFlow));
end
O2conc = O2inAir*AirFlow/1000./totFlow*p/1.3806488e-23/T/1e6;
H2Oconc = WaterFlow/1000./totFlow*vappresw(T)/1.3806488e-23/T/1e6;
SO2conc = SO2Flow/1000./totFlow*SO2BottlePpm*1e-6*p/1.3806488e-23/T/1e6;

H2SO4 = zeros(size(WaterFlow));
for i=1:numel(H2SO4)
    H2SO4(i)=cmd_calib1Matlab(O2conc(i),H2Oconc(i),SO2conc(i),ID/10/2,L/10,Q*1000/60,It,T,p,fullOrSimpleModel);
end
```

Consider here the weighted average of the temperatures displayed on the four MFCs during the experiment

I_t is obtained from an actinometry experiment at a flow rate Q' ; here, it must be scaled to the inlet flow rate Q

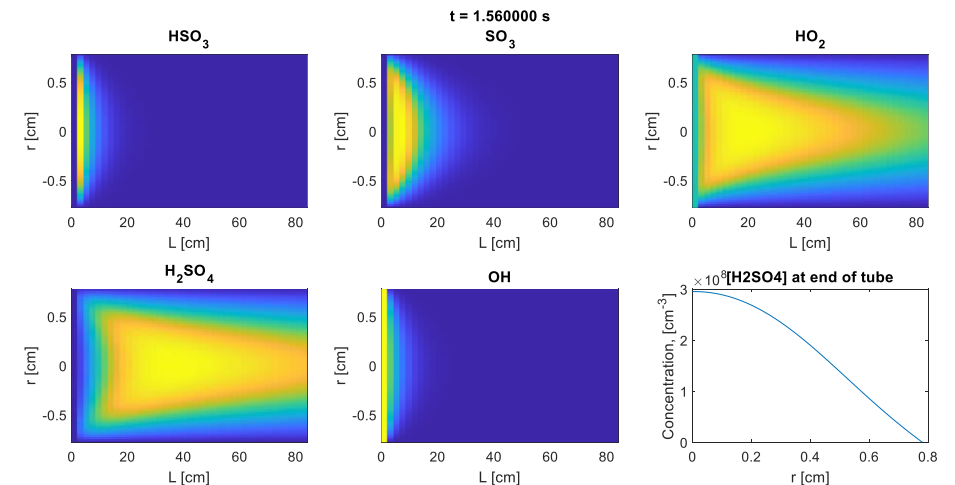
after: setup A
before: setup B

Modeling

Output file (.mat)

- Running `Input_Parameters.m` provides, for each calibration step, the total flow rate together with the modeled concentrations of sulfuric acid (H_2SO_4), water vapor (H_2Oconc), O_2 (O_2conc), and SO_2 (SO_2conc).
- To calculate the calibration factor, only the values contained in the vectors **H2SO4** and **H2Oconc** are required.

Workspace	
Name ▲	Value
AirFlow	70
fullOrSimpleModel	'full'
H2Oconc	[8.7101e+15;1.7209e+16;3.3605e+16;4.9243e+16;6.4174e+16]
H2SO4	[1.7318e+08;3.4646e+08;6.8124e+08;1.0013e+09;1.3072e+09]
i	5
ID	15.6000
It	5.9100e+11
L	840
N2Flow	10
O2conc	[3.4454e+16;3.4037e+16;3.3232e+16;3.2464e+16;3.1731e+16]
O2inAir	0.2090
outflowLocation	'after'
p	98400
Q	10.1200
SO2BottlePpm	5000
SO2conc	[3.5325e+13;3.4897e+13;3.4072e+13;3.3285e+13;3.2534e+13]
SO2Flow	3
T	296.7600
totFlow	[10.1980;10.3230;10.5730;10.8230;11.0730]
WaterFlow	[125;250;500;750;1000]



Data analysis: calculate calibration factor

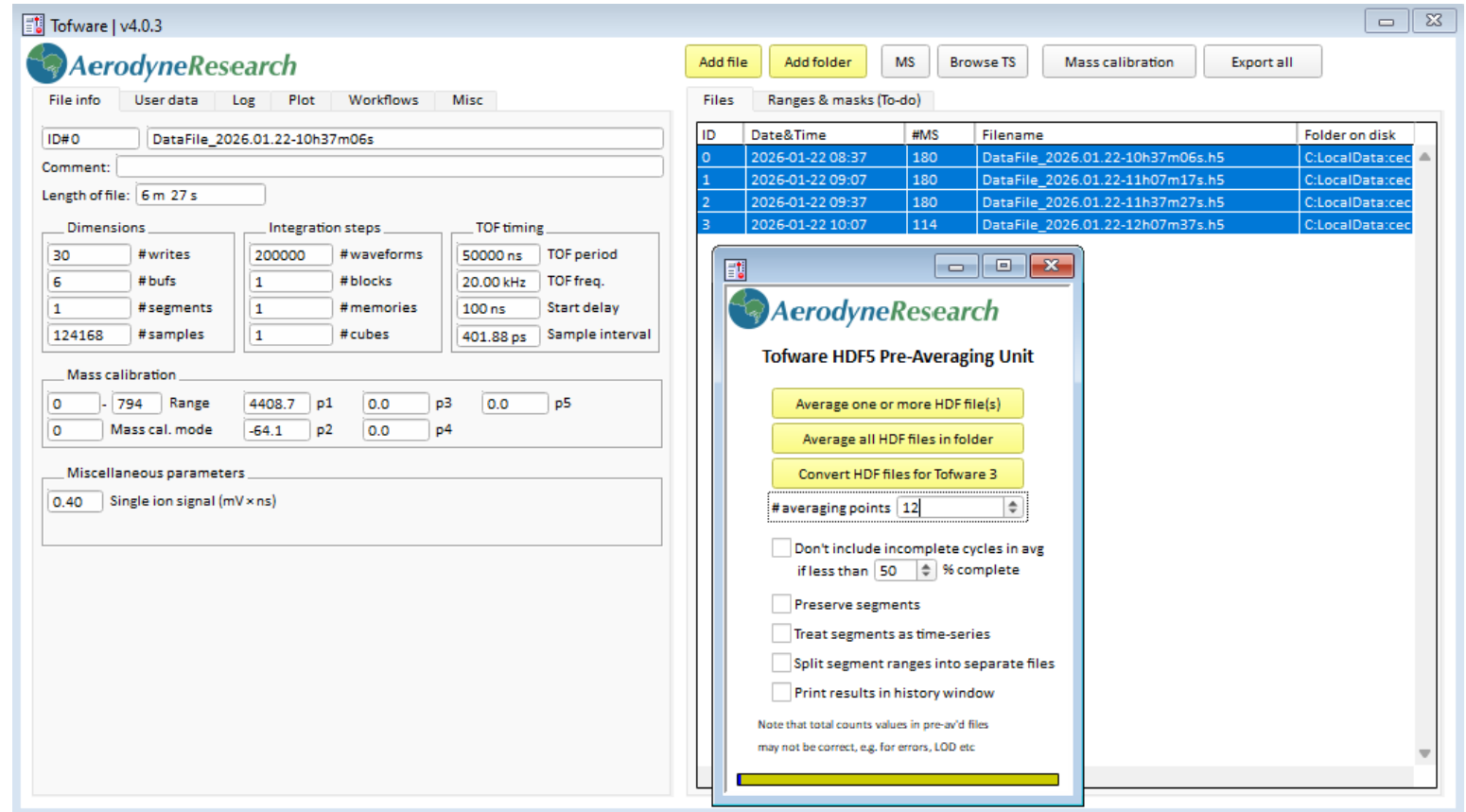
Tofware: data preprocessing
+
Python: calculate calibration factor

TofTools: data preprocessing
+
Matlab: calculate calibration factor

Data analysis: calculate calibration factor

Tofware

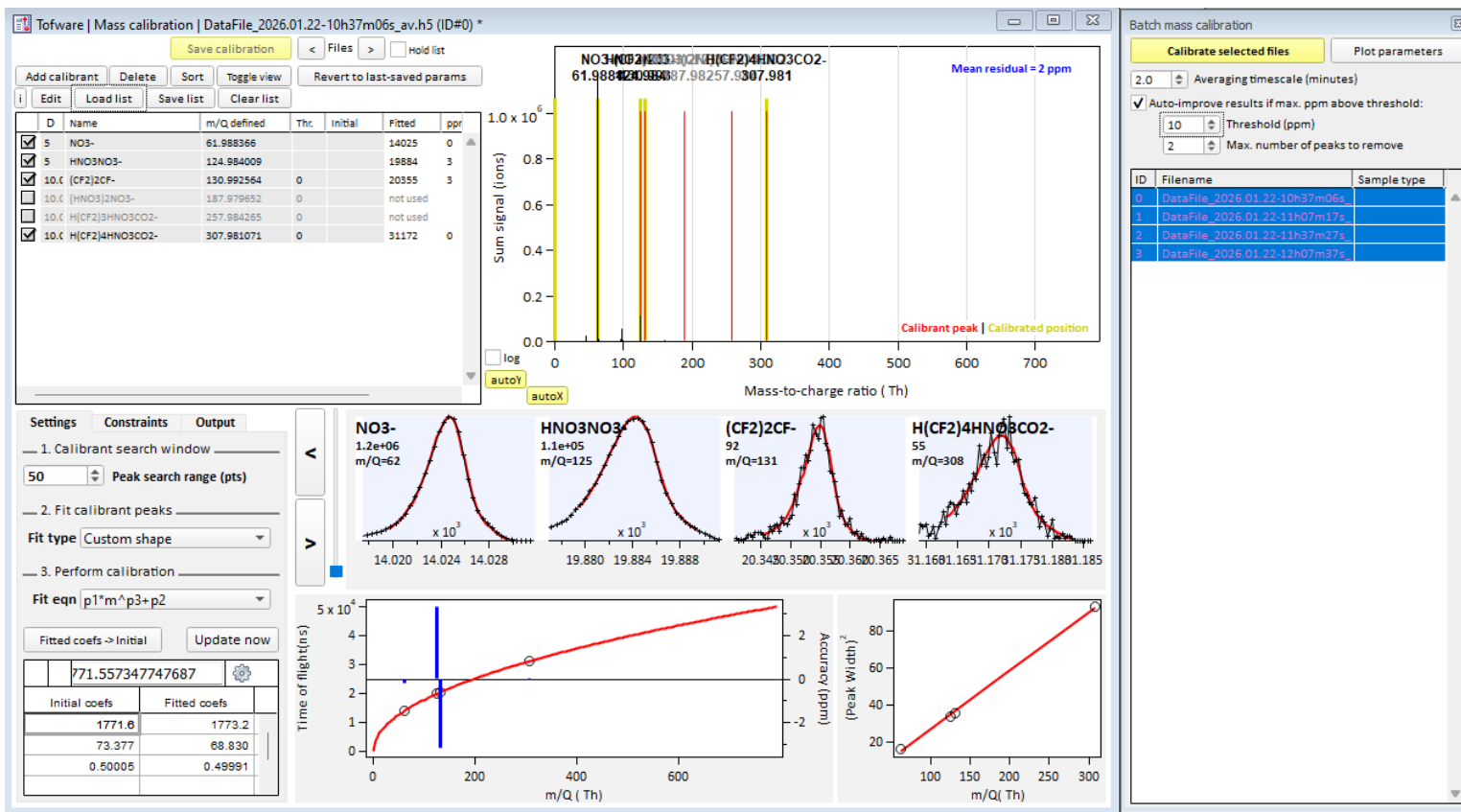
- ❑ Since the calibration steps typically last **10 minutes**, it is recommended to **pre-average the data every 2 minutes** during preprocessing.



Data analysis: calculate calibration factor

Tofware

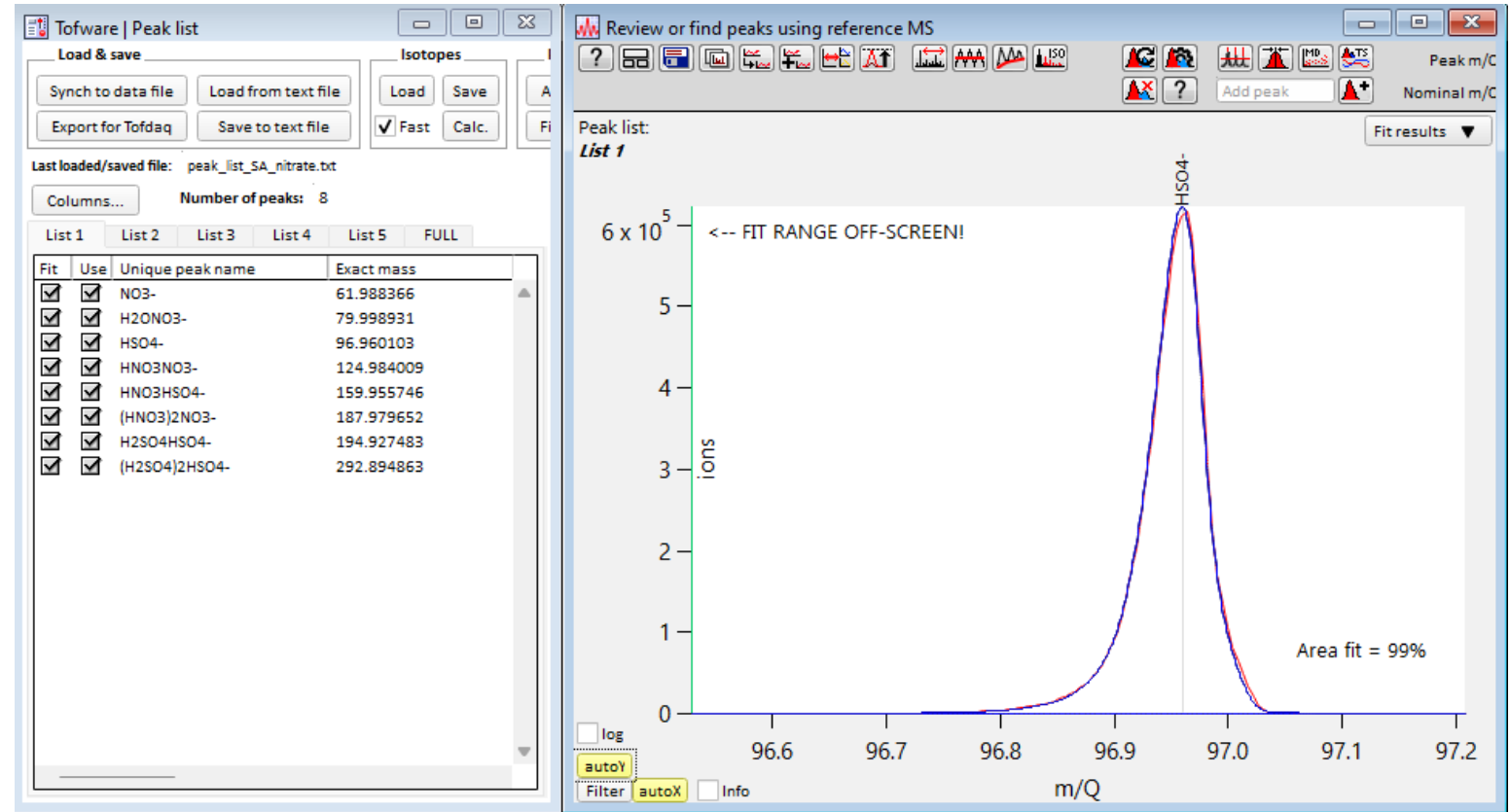
- Calibrants can be chosen among **reagent ions** (e.g., nitrate and its cluster ions in this case) or, preferably, among **stable background peaks**, such as the **PTFE contamination peaks** used in this example.



Data analysis: calculate calibration factor

Tofware

- ❑ The peaks visible in the spectrum should then be fitted so that the sulfuric acid signal $S_{H_2SO_4}$ can be calculated as described in slide 4.



Data analysis: calculate calibration factor

Python

- Use the notebook `calculate_sa_calibration_factor.ipynb` that you can find in [GitHub](#) to compute the calibration factor

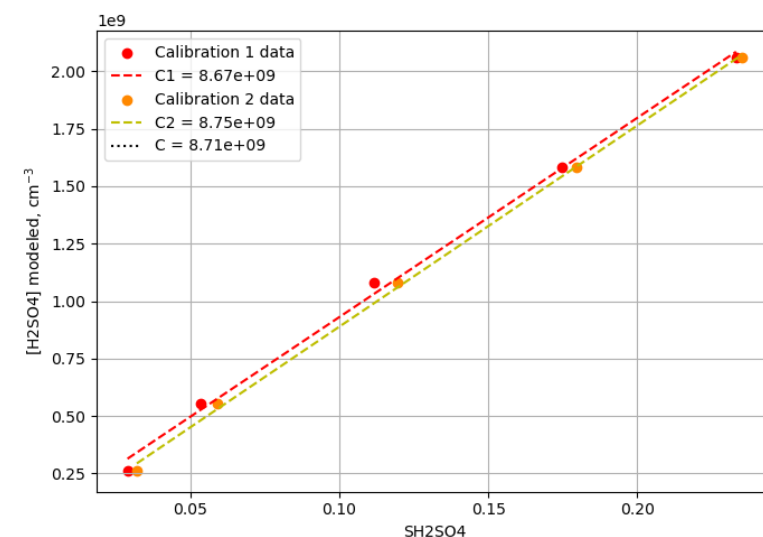
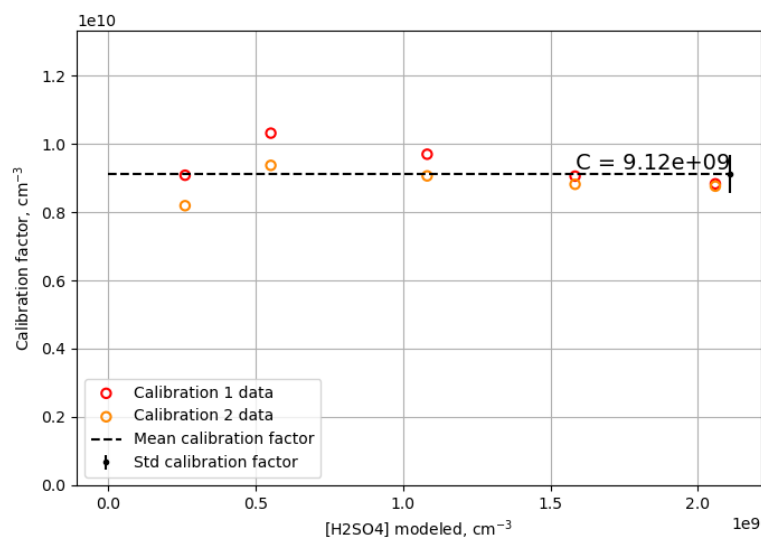
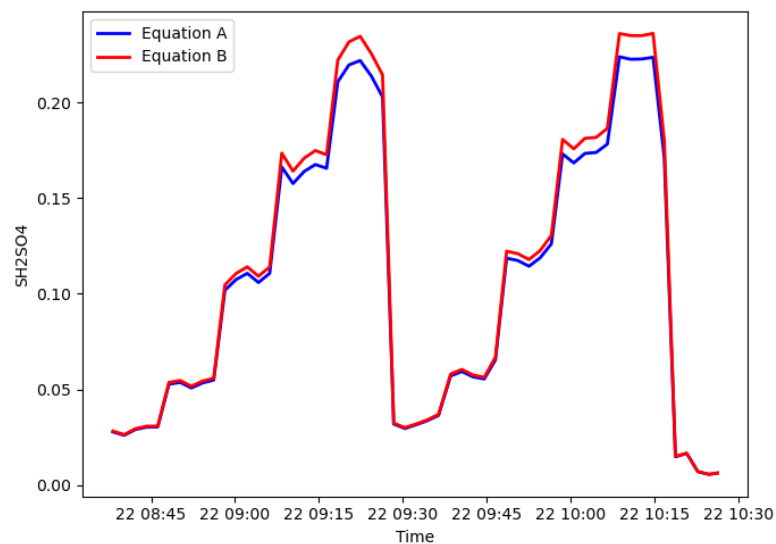
Sulfuric Acid Calibration for CI-API-TOF

Author: Cecilia Righi
Last update: March 2026

Description: This notebook determines the calibration factor of a CI-API-TOF instrument using sulfuric acid calibration experiment data together with modeled sulfuric acid concentrations.

Workflow:

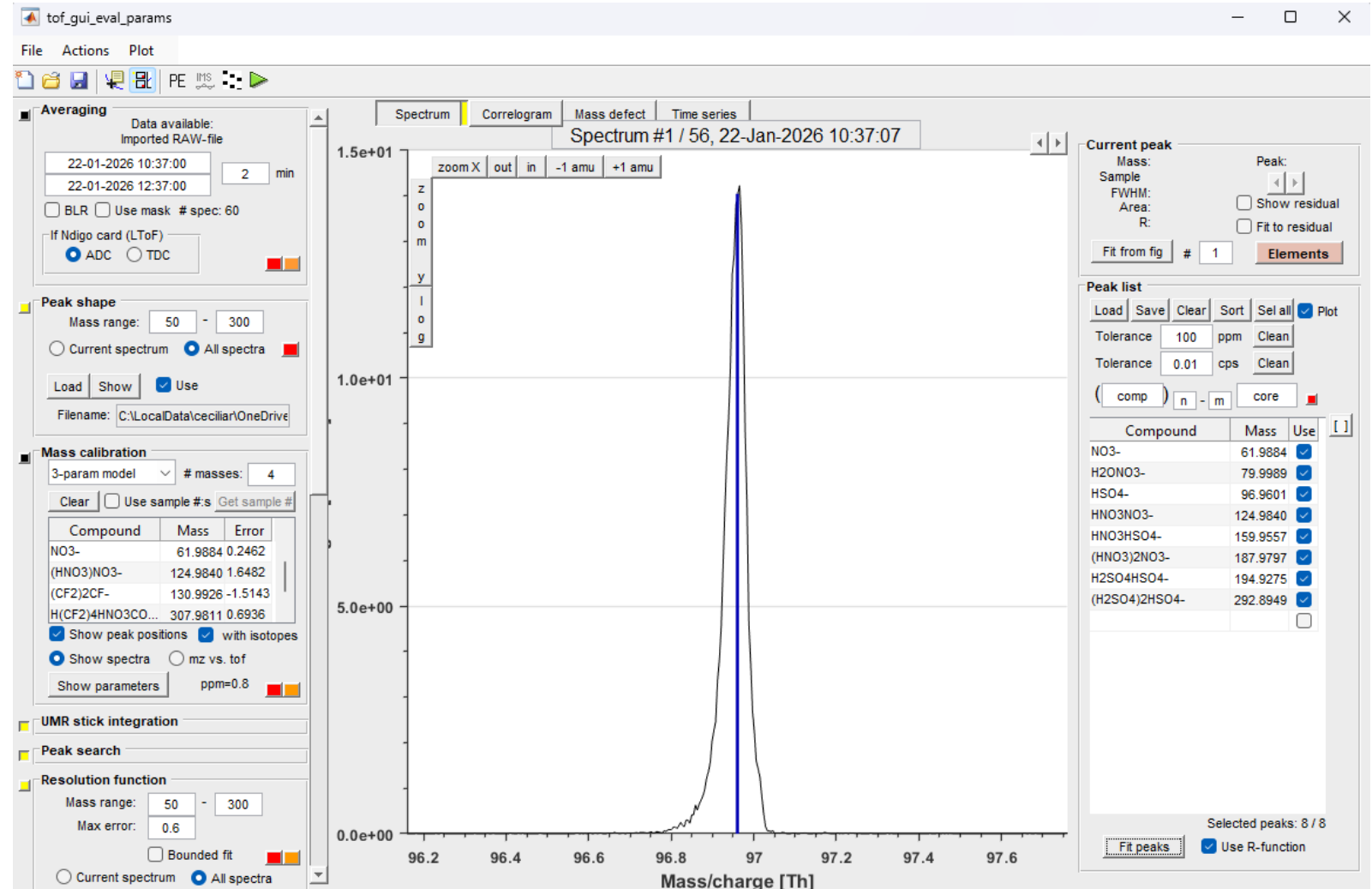
1. Load previously preprocessed CI-API-TOF data from the calibration experiment.
2. Compute the normalized sulfuric acid signal (**SH2SO4**).
3. Calculate the average SH2SO4 value for each calibration step.
4. Load previously modeled H2SO4 concentrations (**[H2SO4]_modeled**).
5. Determine the calibration factor as:
 - **C_avg**: the average of the calibration factor calculated for each step
 - **C_lin**: the slope of the linear regression $[H2SO4]_{modeled} = f(SH2SO4)$



Data analysis: calculate calibration factor

TofTools

- ❑ All recommendations provided for *Tofware* preprocessing (including pre-averaging settings, calibrant selection, and peak list definition) also apply when preprocessing the data using *TofTools*.

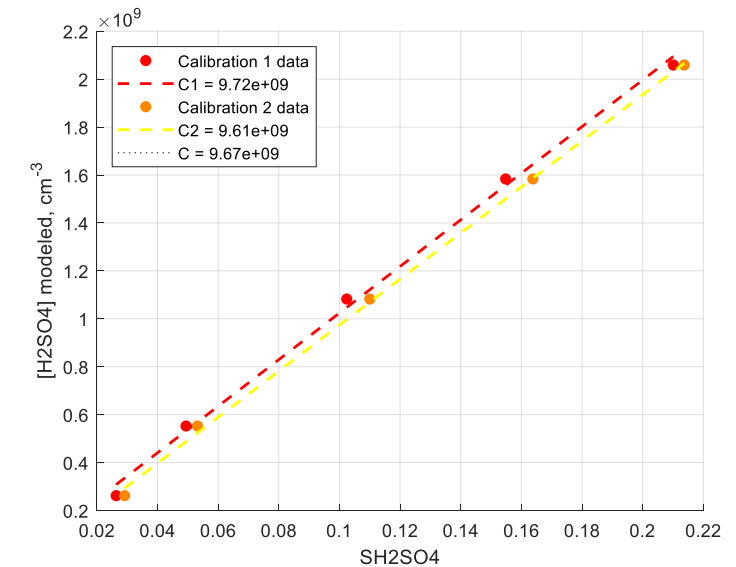
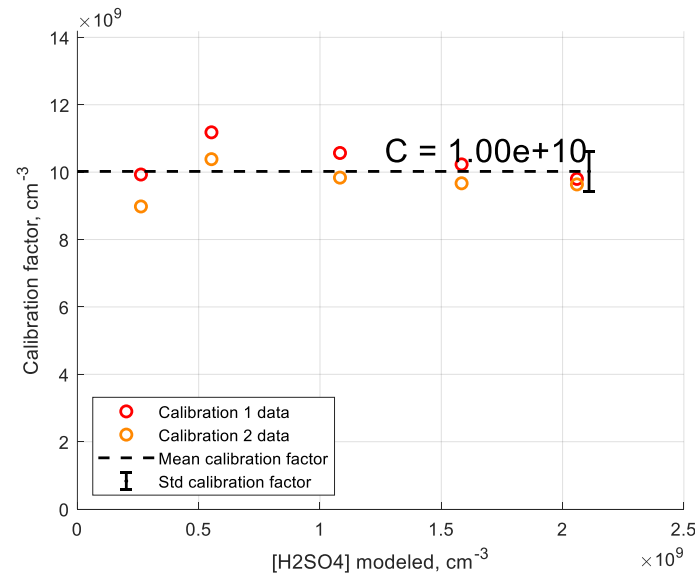
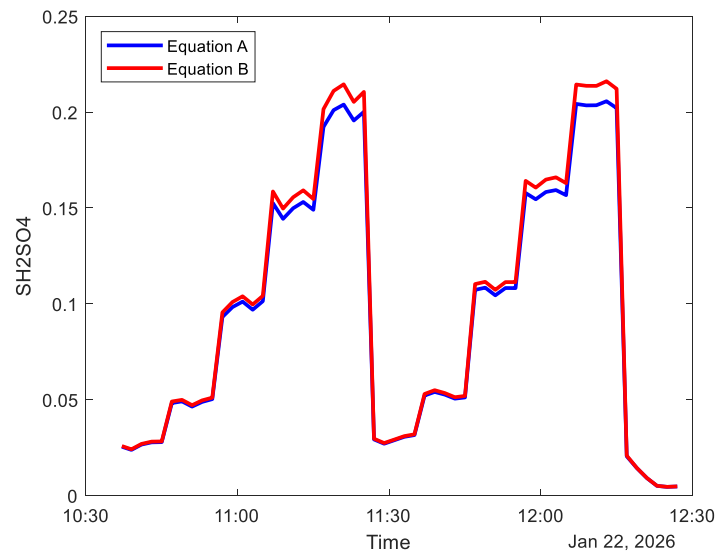


Data analysis: calculate calibration factor

Matlab

- ❑ Use the script `calculate_sa_calibration_factor.m` that you can find in [GitHub](#) to compute the calibration factor

```
calculate_sa_calibration_factor.m  avg_from_fig.m  +
1  %% Sulfuric Acid Calibration for CI-API-TOF
2  %
3  % Author: Cecilia Righi
4  % Last update: March 2026
5  %
6  % Description:
7  % This script determines the calibration factor of a CI-API-TOF instrument
8  % using sulfuric acid calibration experiment data together with modeled
9  % sulfuric acid concentrations.
10 %
11 % The workflow includes:
12 % 1. Load previously preprocessed CI-API-TOF data from the calibration experiment.
13 % 2. Compute the normalized sulfuric acid signal (SH2SO4).
14 % 3. Calculate the average SH2SO4 value for each calibration step.
15 % 4. Load previously modeled H2SO4 concentrations ([H2SO4]_modeled).
16 % 5. Determine the calibration factor as:
17 %     - C_avg: the average of the calibration factor calculated for each step
18 %     - C_lin: the slope of the linear regression [H2SO4]_modeled = f(SH2SO4)
19 %
20
21  %% -----
```



Data analysis: calculate calibration factor

C_{avg} vs C_{lin}

The SA calibration factor, C_{SA} , can be calculated as

A) mean of individual $C_{SA,i}$:

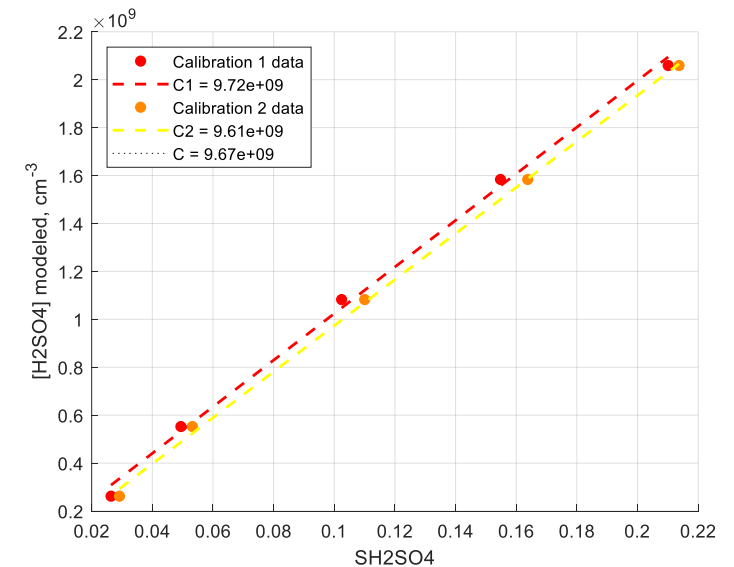
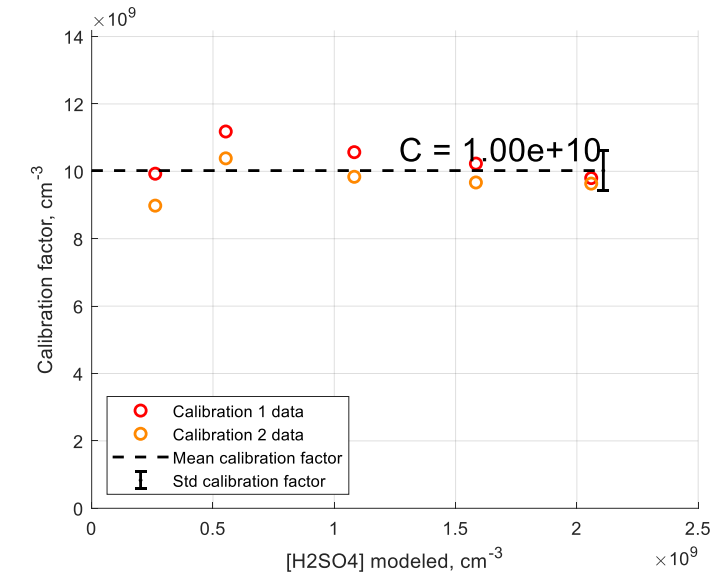
$$C_{avg} = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{x_i} = \frac{1}{n} \sum_{i=1}^n C_{SA,i}$$

where $y_i = [H_2SO_4]_{modeled,i}$
 $x_i = S_{H_2SO_4,i}$

B) slope of the linear regression fit ($y_i = m(x_i) \cdot x_i$):

$$C_{lin} = m = \frac{cov(x,y)}{var(x)}$$

where $cov(x,y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$
 $var(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$



Data analysis: calculate calibration factor

C_{avg} vs C_{lin}

When the calibration factors $C_{SA,i}$ computed for each step i of the calibration experiment lie on a same horizontal line:

$$y_i = m(x_i) \cdot x_i,$$

$$\text{with } m(x_i) = \frac{y_i}{x_i} = \text{constant} = m \text{ and } C = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{x_i} = m$$

On the contrary, if the calibration factors $C_{SA,i}$ do not lie on the same horizontal line:

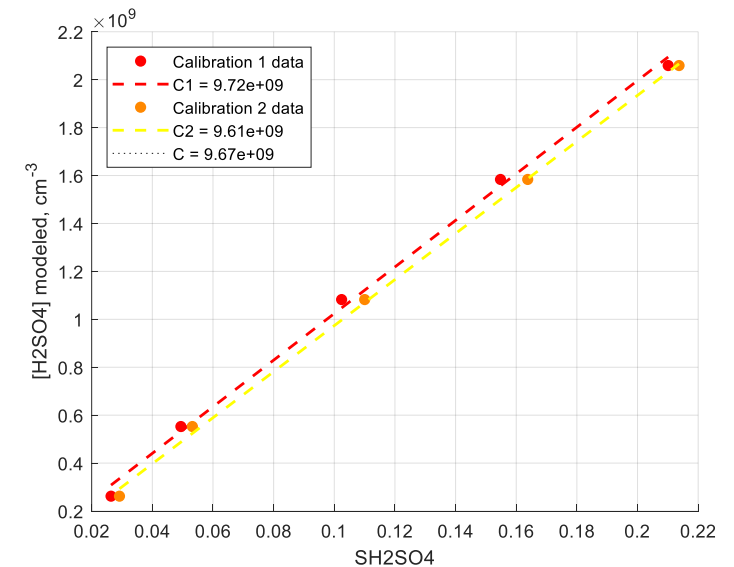
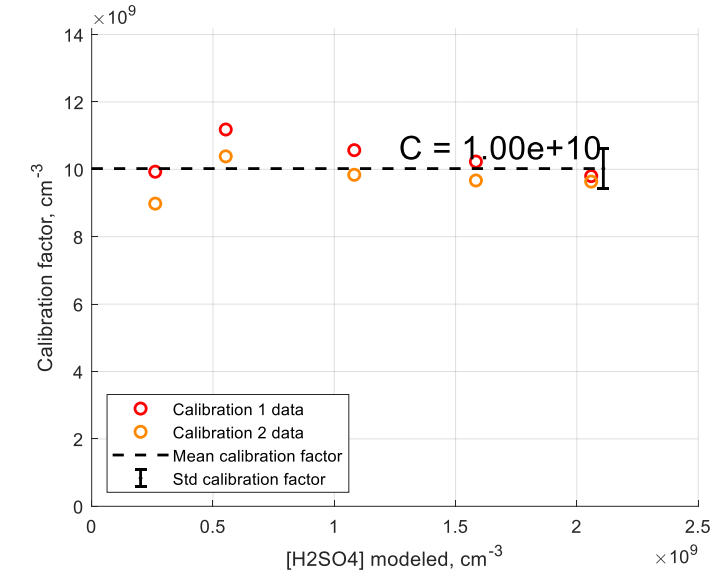
$$m(x_i) = \frac{y_i}{x_i} \neq \text{constant} \text{ and } C \neq m$$

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \text{ gives more importance to points further from the mean signal}$$



When $C_{SA,i}$ increases with x_i , $C_{lin} > C_{avg}$

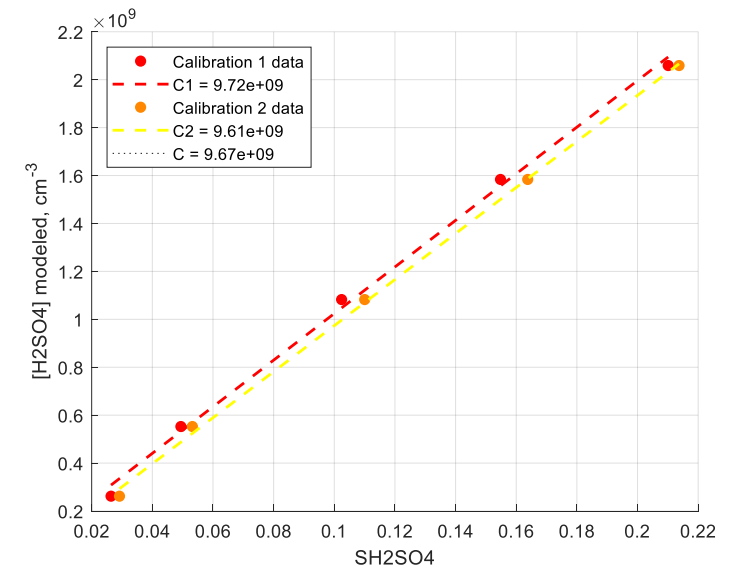
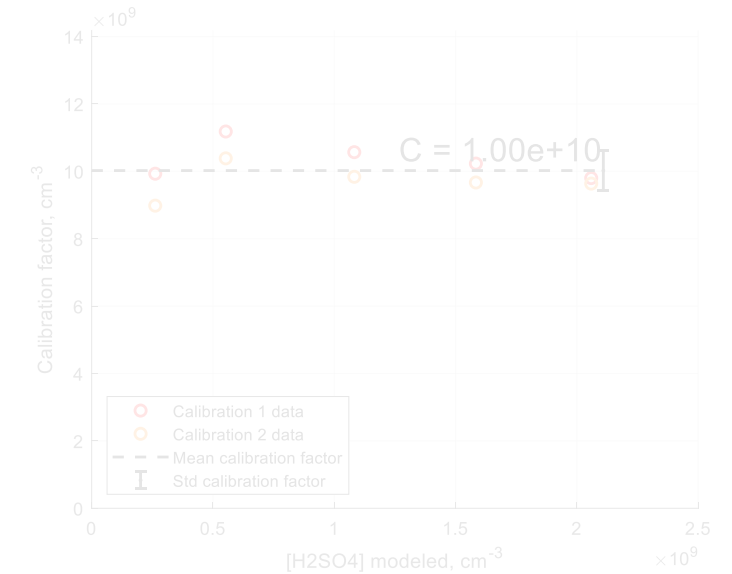
When $C_{SA,i}$ decreases with increasing x_i , $C_{lin} < C_{avg}$



Data analysis: calculate calibration factor

C_{avg} vs C_{lin}

We recommend using the calibration factor obtained from the slope of the linear regression (Method B).



Data analysis: calculate calibration factor

C_{avg} vs C_{lin}

Method A plot remains useful for quickly assessing whether the calibration steps are stable or if there is a visible increasing trend in $C_{SA,i}$ (see slide 12).

We recommend using the calibration factor obtained from the slope of the linear regression (Method B).

